

Basic Principles of Medicine 2
Module: DM
Lecture No: 25

Title of lecture
Disorders of amino acid metabolism

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Objectives: Inherited disorders of amino acid metabolism

SOM. MK.I.BPM2.3.DM.2.BCHM.1230	Analyze deficiencies in biochemical pathways that are associated with inborn errors of metabolism.
SOM. MK.I.BPM2.3.DM.2.BCHM.1231	Discuss the appropriate use of language when referring to patients who have disorders with intellectual disability and delivery of findings to parents.
SOM. MK.I.BPM2.3.DM.2.BCHM.1232	Describe laboratory findings that are diagnostic for inborn errors of amino acid metabolism.
SOM. MK.I.BPM2.3.DM.2.BCHM.1233	Outline the catabolic pathway of phenylalanine and tyrosine and associated disorders like Phenylketonuria (Classic-PKU I and tetrahydrobiopterin deficiency – PKU II, maternal PKU), Alkaptonuria.
SOM. MK.I.BPM2.3.DM.2.BCHM.1234	Differentiate between the inborn errors of amino acid metabolism based on clinical features, laboratory findings, and vitamin/ coenzyme supplementation.
SOM. MK.I.BPM2.3.DM.2.BCHM.1235	Outline the catabolic pathway of branched chain amino acids and explain the metabolic basis for Maple syrup urine disease, Methylmalonic aciduria.
SOM. MK.I.BPM2.3.DM.2.BCHM.1219	Analyze and correlate clinical features and laboratory findings to the metabolic basis of cystinuria and Hartnup's disease
SOM. MK.I.BPM2.3.DM.2.BCHM.1236	Outline the catabolic pathway of methionine and cysteine and outline the two possible fates of homocysteine. Explain metabolic basis for homocystinuria.

Recommended Reading – Lippincott's Illustrated Reviews – 9th edition

Lippincott's Biochemistry 9th Edition

<https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/book.aspx?bookid=3402>

Chapter 20. Amino Acids: Degradation and Synthesis Pg. 290

Text Questions 20.1-20.7 pg 306

[Amino Acids: Degradation and Synthesis | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#)

Chapter 19. Amino Acids: Nitrogen Disposal pg. 271

Text Questions 19.1-19.5 pg. 288

[19. Amino Acids: Nitrogen Disposal | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#)

Chapter 21. Amino Acids: Conversions to Specialized Products pg. 308

Text Questions 21.1-21.6 pg. 322

Concept map: Pg 321

[Amino Acids: Conversion to Specialized Products | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#)

Things to focus on:

- Enzymes deficiencies and if they require coenzymes (vitamins)
- Clinical features or basis if described in the notes
- Therapies if available
- See tables at the end of the lecture notes to guide your studies

Amino Acid Metabolism:

Inherited disorders of amino acid catabolism

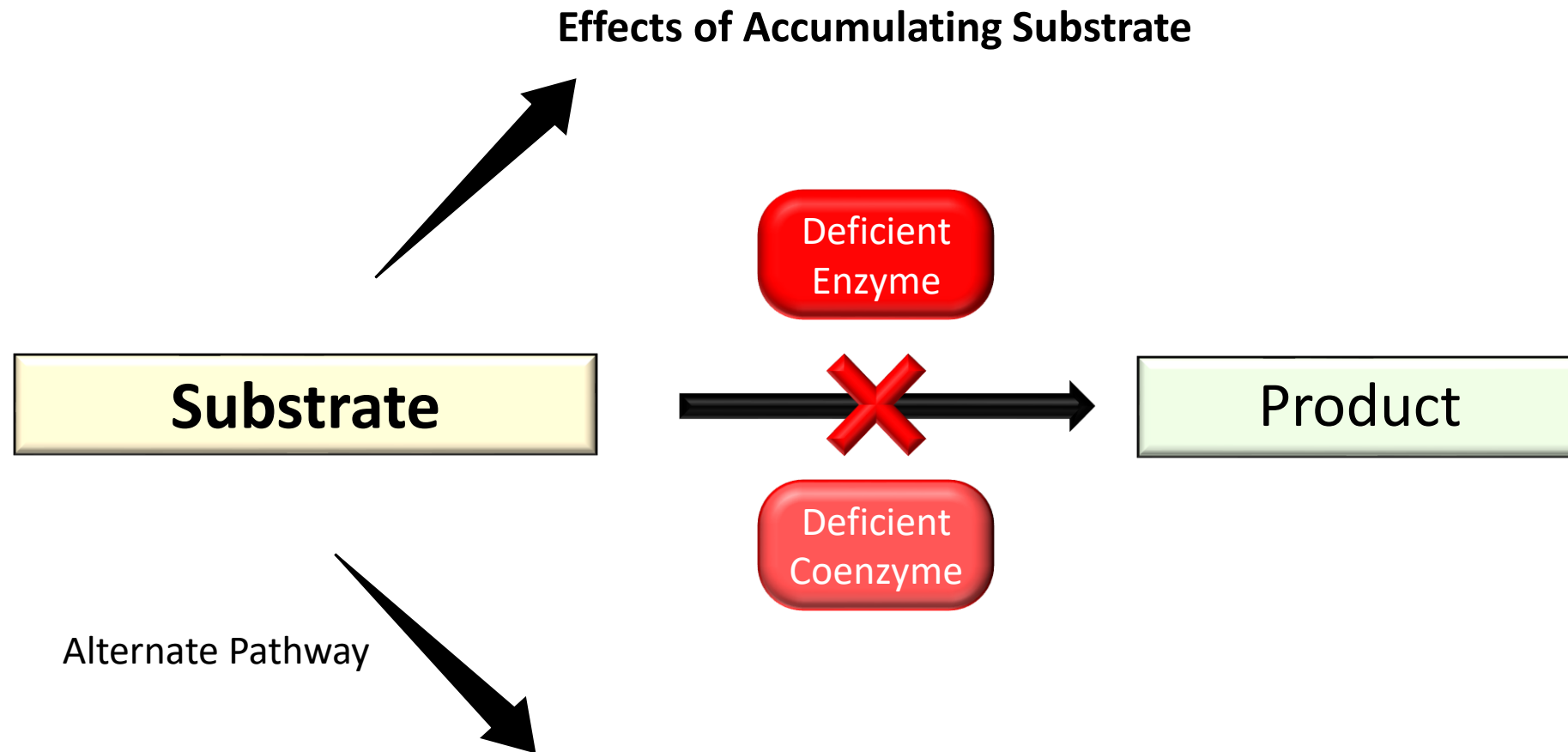


- Enzyme deficiencies involved in amino acid metabolism result in accumulation of toxic substances
 - Greater enzyme deficiencies = increased toxicity
- Brain, liver and kidneys most affected
- Acute symptoms typically during catabolic state
- Most aminoacidopathies are identified by newborn screening

“The day after being discharged from the hospital the parents of a new-born baby receive an emergency phone call telling them to not feed their baby, but to report to their physician immediately!”

Link this concept: Recall

General overview of biochemical changes in a typical enzyme deficiency



**Alternate Products may be Toxic and
Contribute to Pathology Effects**

Amino acid metabolism

Disorders of catabolism: specific amino acids

Phenylalanine and Tyrosine

- Phenylketonuria, albinism, alkaptonuria

Branched chain amino acids (Valine, isoleucine, leucine)

- Maple syrup urine disease, methylmalonic aciduria

Sulfur containing amino acids (Methionine and cysteine)

- Homocystinuria

Transport Defects

- Cystinuria and Hartnup

PKU Scenario

- 'June 24, 1996, was the proudest day of my life. It was the day of my son Z's long-awaited and much-anticipated arrival, albeit a month earlier than we expected. Everything went smoothly and after a brief re-admission for dehydration from poor feeding related to his prematurity, we were back at home enjoying being parents.'
- When Z was a week old, we received a call from his pediatrician, who is also a friend and mentor, informing us that his **newborn screen revealed an elevated phenylalanine level.**
- My immediate reaction was one of panic because I realized that I did not have any idea what it really meant to have PKU. All I remembered was some peripheral reference to it during **medical school biochemistry.**'

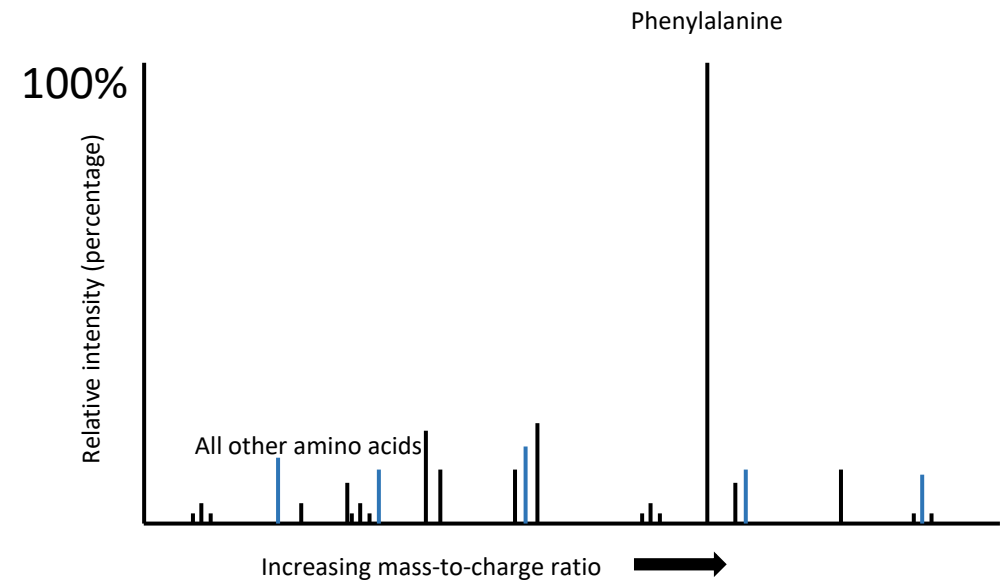
Newborn Screening



- Amino acid concentration in blood quantitated from blood spot
- Other organic molecules can also be quantitated from blood spot
- Molecular profile based upon the mass / charge ratio of the fractured molecule



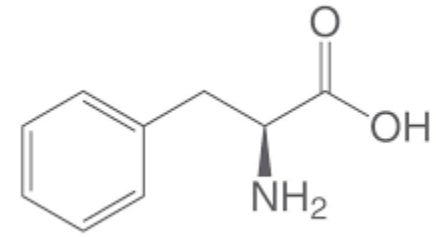
The Mass Spectrometer



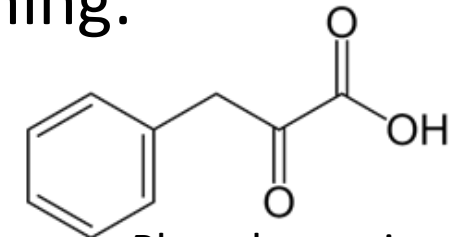
Phenylalanine Metabolism

Phenylketonuria (PKU)

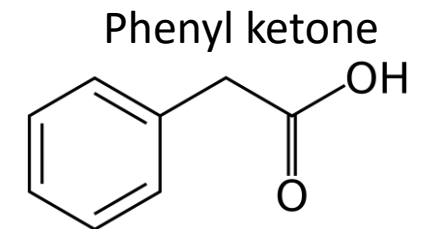
- Relatively common of the **rare** disorders
 - Autosomal recessive disorder
- Responds very well to dietary treatment
- Detected during the first week of life by neonatal screening:
 - The Guthrie test (historical)
 - Mass Spec (current use)
- If test is positive
 - Immediately - place infant on low Phenylalanine (Phe) diet
 - Then - test again, in case of false positive
- Some centers test infants twice
 - First test might be false negative due to maternal clearance



Phenylalanine



Phenyl pyruvic acid



Phenyl ketone

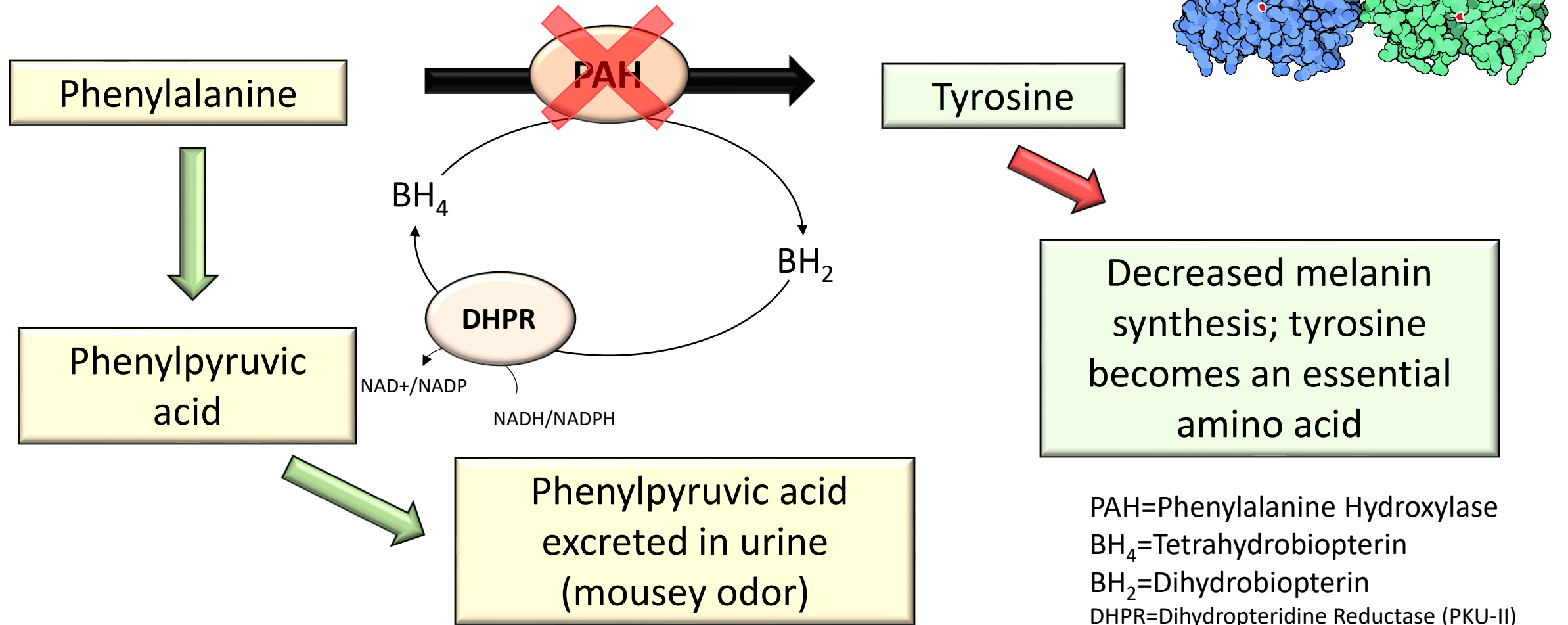
Phenylketonuria (PKU)



- Children confirmed to have PKU, have to be put on a low Phe diet
 - Medical food that is modified to have low Phe
 - Low protein diet
 - Avoid aspartame (artificial sweetener; i.e., Diet Coke, etc)
- Dietary restriction of Phe to be started immediately on diagnosis
- Blood Phe levels have to be frequently monitored
- If treated, children lead a normal life
 - Except for the dietary restrictions
- Recommended lifelong adherence, but after development, patients often stray from the diet (not recommended)
- First successfully treated inborn error of metabolism by dietary restriction was in the 1950s (Bickel, 1953)

Biochemical defect in PKU

PKU I (Classic PKU)

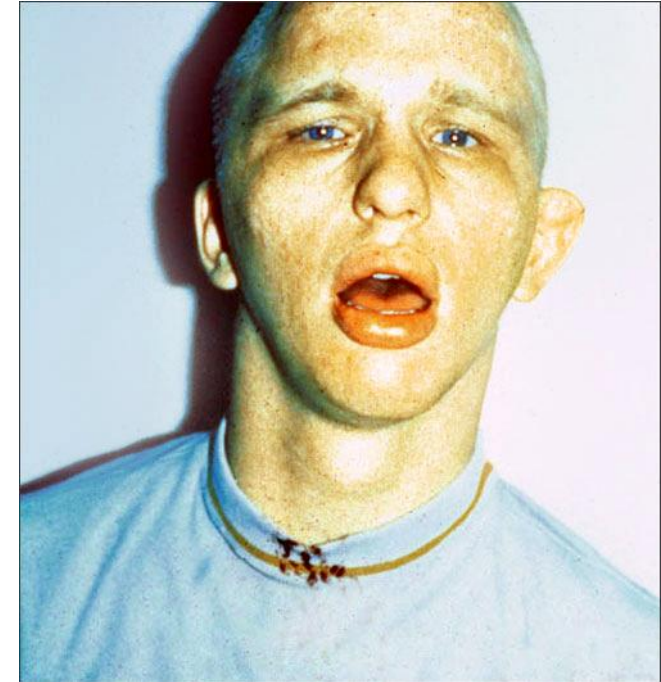


Types of PKU

- Classic PKU-
 - mutation in phenylalanine hydroxylase
- Malignant PKU (PKU-II)-
 - mutation in restoring BH_4 from qBH_2
 - *de novo* synthesis of BH_4
- Maternal PKU-
 - Female with classic PKU with high blood phenylalanine (Phe) levels

Characteristics of PKU I (classic PKU) in Untreated Patients

- CNS symptoms: child has a delay in milestones and a low IQ (if not treated immediately). Seizures if blood Phe are higher than the recommended level
- Phe is metabolized to phenylpyruvate, phenylacetate & phenyllactate (excreted in urine) resulting in mousey odor of urine
- Phe levels in blood are elevated
- Decreased pigmentation of skin and hair as tyrosine conversion to melanin is inhibited by the elevated Phe levels (inhibits tyrosinase)



Most untreated PKU patients have lighter coloring than parents/siblings from reduced melanin synthesis

PKU Before Treatment was known



- Intellectual disability
- Seizures
- Spasticity
- Autistic behaviors
- Hypopigmentation
- Skin rashes

Tested at 2 yrs old then treated



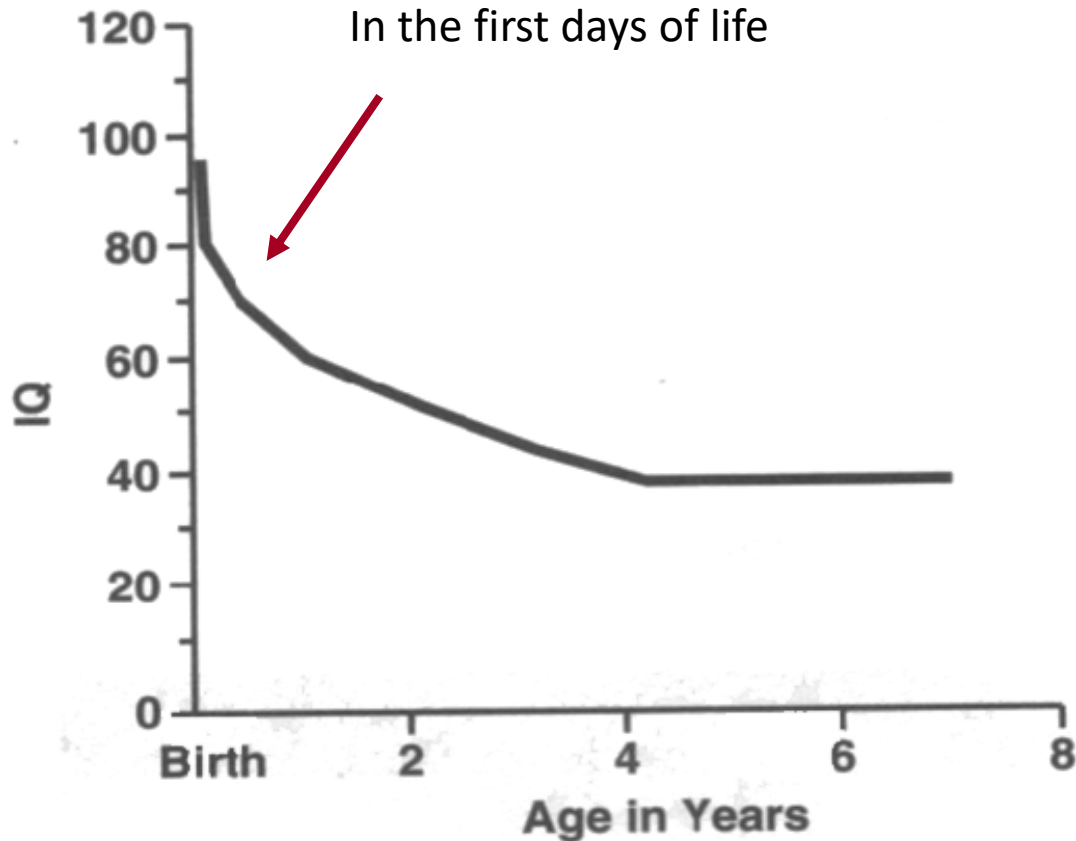
Siblings with PKU – the sibling treated at birth develops normally

Tested 1 day old: treated by dietary restriction of phenylalanine

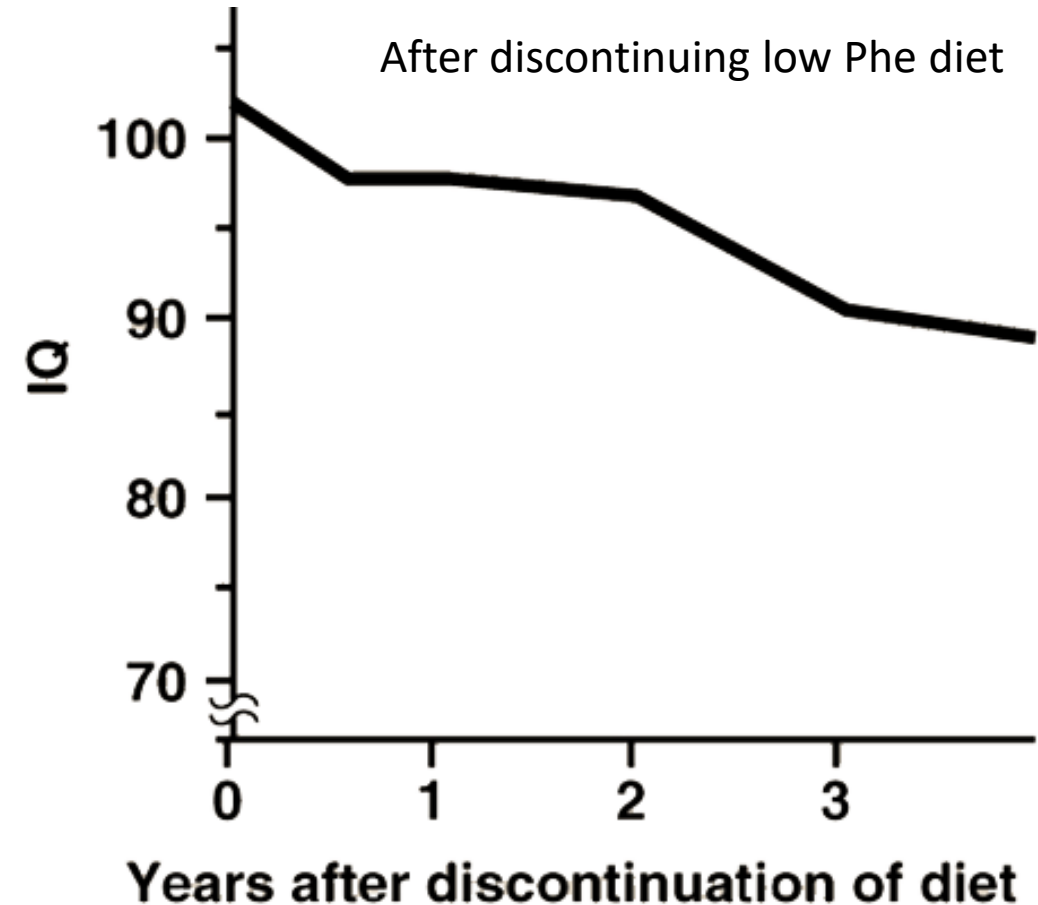


- First successfully treated inborn error of metabolism by dietary restriction was in the 1950s (Bickel, 1953)

Reduction in IQ is seen if Phe Levels are not managed in PKU Patients



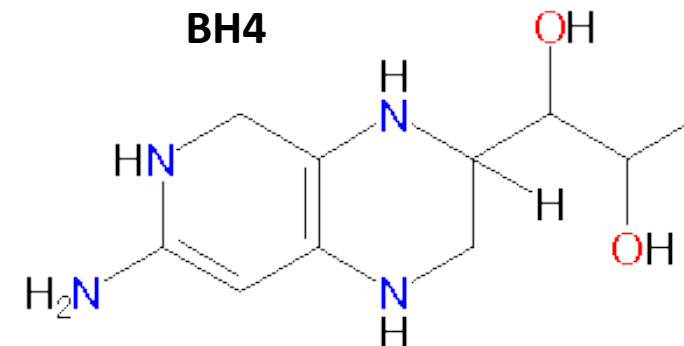
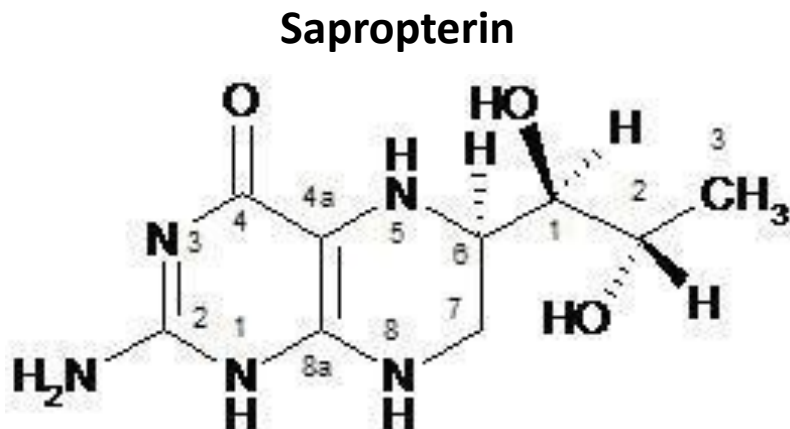
- Most severe IQ decline is in first few days
- Early screening and intervention is essential



- Teenagers often become noncompliant with diet
- Females planning a pregnancy should follow very strict diet months before conception

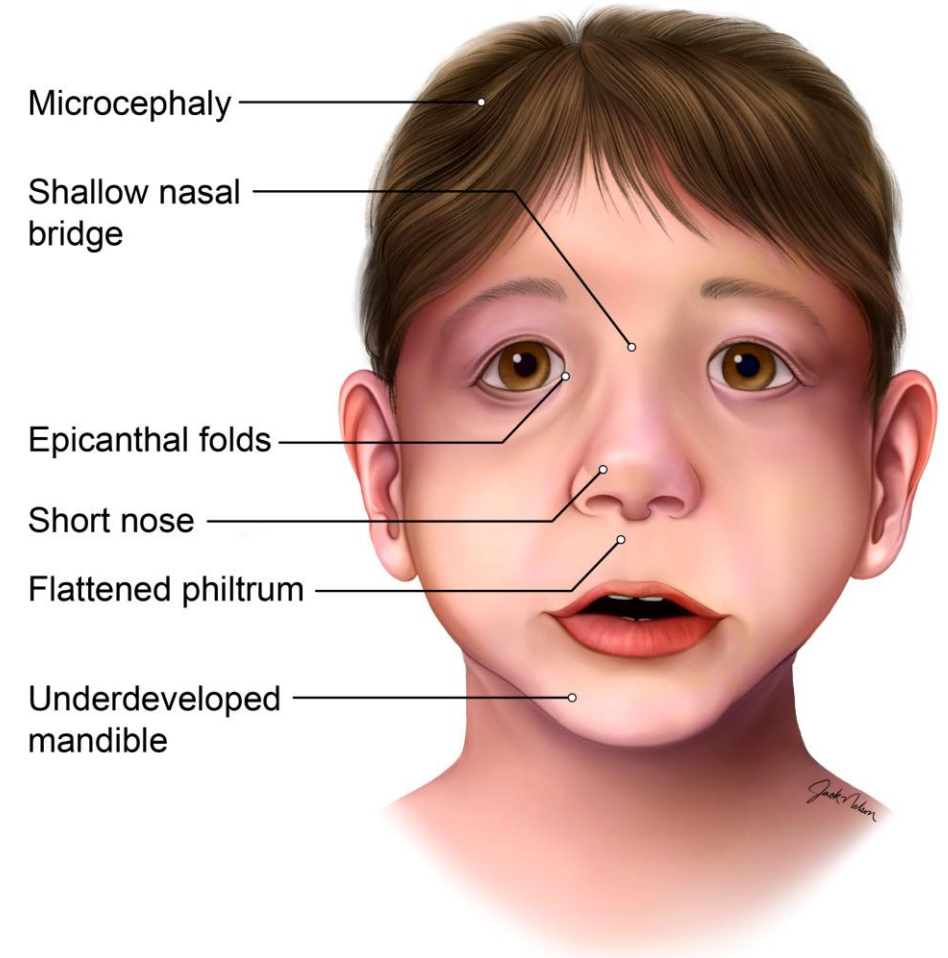
Treatment of PKU

- **Avoidance** – dietary treatment – reduction in phenylalanine (careful monitoring of Phe intake during growth of the patient)
- **Sapropterin** (synthetic form of BH_4) may help some PKU patients
 - Typically mild or moderate forms of the disease
 - These patients may have a mutant form of the enzyme that has a lower affinity for the cofactor qBH_2/BH_4
 - Considered the first non-dietary treatment for PKU
- **Enzyme replacement therapy- new! Phenylalanine ammonia lyase**
 - Patients can inject themselves every day, FDA approved



Maternal PKU syndrome

- Women with PKU must maintain **low** Phe levels before conception and during pregnancy.
 - Strict adherence must be maintained prior to conception and throughout pregnancy
- **High maternal blood Phe leads to fetal defects:**
 - Microcephaly
 - Lack of mental development, severe neurological problems
 - Congenital heart defects
 - Facial dysmorphism- long philtrum, epicanthal folds, micrognathia, short nose, microphthalmia, flat midface, facies similar to fetal alcohol syndrome
- Elevated phenylalanine has **teratogenic** properties



Note: Maternal PKU syndrome occurs even though the fetus is **NOT** deficient in phenylalanine hydroxylase (Typically the fetus is heterozygous)

Link this concept: Most people are homozygous for normal alleles of PAH gene

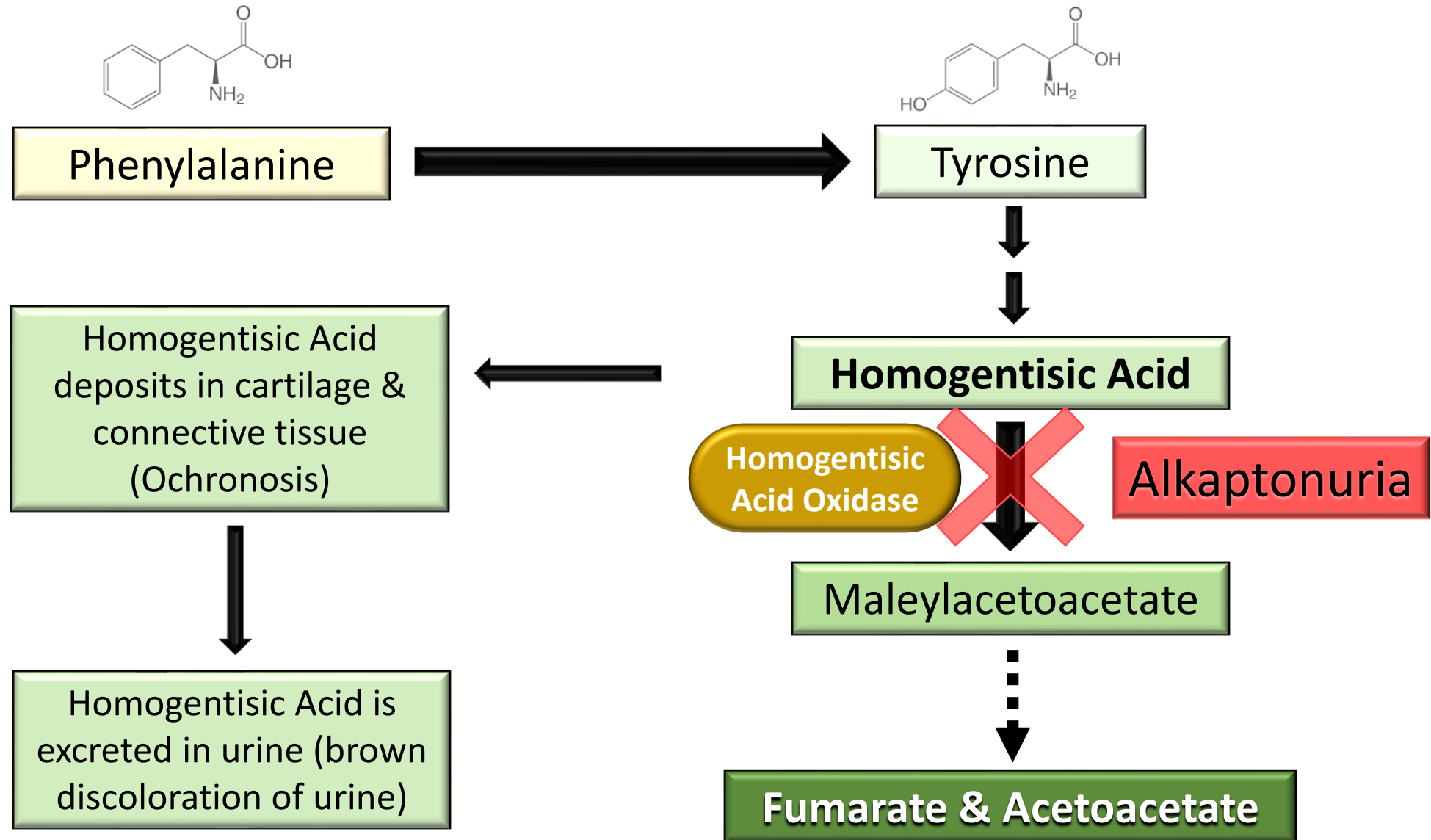
		Mother with PKU I	
		pah/pah	pah/pah
Father from the population	PAH/PAH	PAH/pah +/-	PAH/pah +/-
	PAH/PAH	PAH/pah +/-	PAH/pah +/-

Offspring of a person with PKU

All children, once born, will be normal (heterozygous). But during fetal development, the high circulating levels of maternal serum PHE would be teratogenic if the mother is not under strict dietary control of phenylalanine consumption.

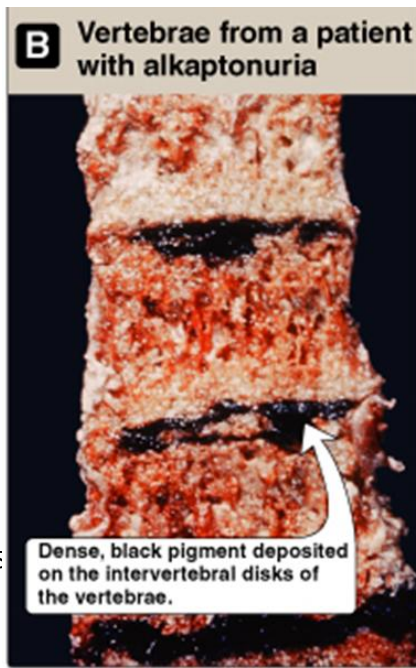
Tyrosine Metabolism

Tyrosine Metabolism: Alkaptonuria

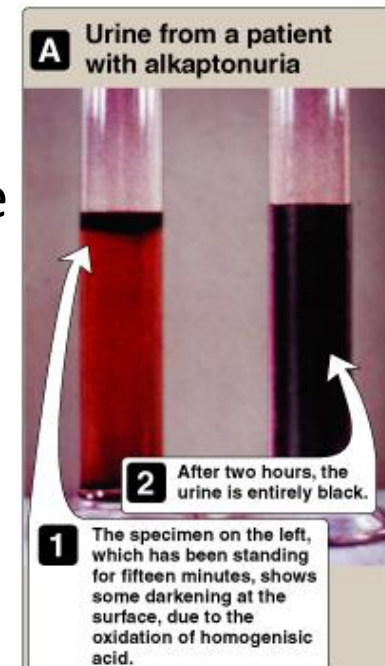


Tyrosine Metabolism: Alkaptonuria

- Relatively benign inborn error of phenylalanine - tyrosine catabolism
- Deficiency in Homogentisic Acid Oxidase
- Homogentisic acid accumulates, may lead to severe arthritis later in life
- Dietary restriction of Phe & Tyrosine may reduce deposition of homogentisic acid

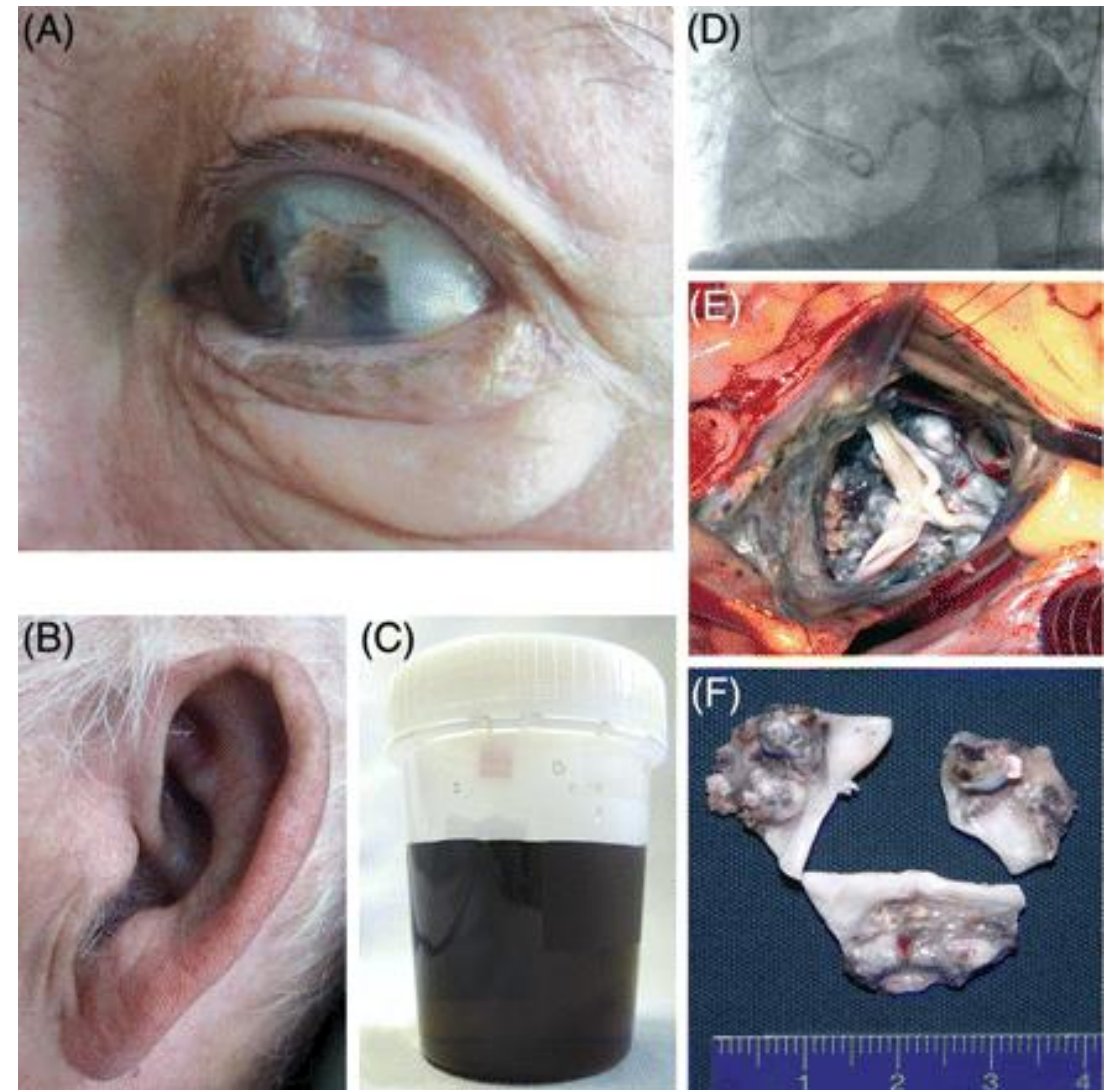


- Darkening of urine on standing
- Discoloration of cartilage and connective tissue
 - Ochronotic pigments (ochronosis)
- Often leads to severe arthritis
- Due to oxidation of excess homogentisic acid



Alkaptonuria scenario

A 71-year-old male patient was referred for the evaluation of a high-grade **aortic stenosis**. His medical history revealed prosthetic hip and joint replacement due to **degenerative arthritis** at the age of 55. On physical examination, kyphosis, and a **bluish-black discoloration of the sclera and auricles** were noted (Panels A and B). The patient reported that his **urine darkens since childhood** (Panel C). Cardiac catheterization demonstrated a severely calcified aortic stenosis. On the basis of the bluish-black discoloration of collagenous tissue (sclera, auricles, aortic valve), history of degenerative arthropathy, and blackening of the urine on standing, diagnosis of alkaptonuria was made and confirmed by increased urinary excretion of homogentisic acid



Link this concept: Recall



Osteogenesis Imperfecta – blue sclera
Child



Alkaptonuria – bluish black discoloration of sclera
Adults

Phornphutkul et al. 2002, N Engl J Med; 347:2111-2121

Compare eye findings from different genetic disorders

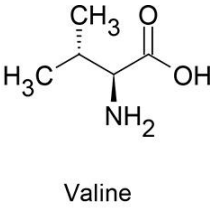
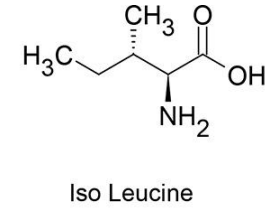
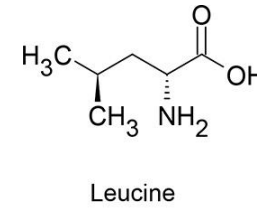
Branched Chain Amino Acid Metabolism

Maple Syrup Urine Disease (MSUD) Scenario

- A child was admitted on the second day of life for observation and did well until breathing problems developed on the third day. The baby became **irritable**, had peculiar cries and stiffened. The peculiar **urinary odor** was noticed on day six. The mother had noticed that smell on two of her previous newborns who died soon after birth, and said it smelled like **maple syrup**.
- The neurologic symptoms progressed very rapidly. The baby became **stiff, unable to suck**, could not breathe, and finally **seized**. He then developed convulsions and pneumonia and died at 14 days of age. Doctors could do nothing for him.
- Not all of the children in this family showed the symptoms.
- Disorders of breathing were the first signs noted and were associated with brain stem swelling.

First reported case of MSUD was in 1945 by Dr. Menkes

Maple Syrup Urine disease (MSUD)



- Inborn error of metabolism: branched chain amino acids
- Mild to severe forms, relative to residual enzyme activity
- Symptoms develop in **neonates aged 3-5 days** & must be treated immediately
- Can be detected early in life by neonatal/newborn screening
- **Severe Form:** Presents with poor feeding, vomiting, poor weight gain and increasing lethargy.
 - **Neurological signs** (e.g., alternating muscular hypotonia and hypertonia, seizures, encephalopathy) develop rapidly.
 - **Ketosis** and the characteristic odor of **maple syrup in the urine** are usually present when the first symptoms develop.
 - Coma and death of the child in early infancy if not recognized and treated
- **Mild Form:** Psychomotor developmental delay, progressive neurological disease, recurrent ketoacidotic episodes

Maple Syrup Urine disease (MSUD)



1-day-old, characteristic opisthotonic position



3-week-old, hypotonia



Very rigid with dermatitis

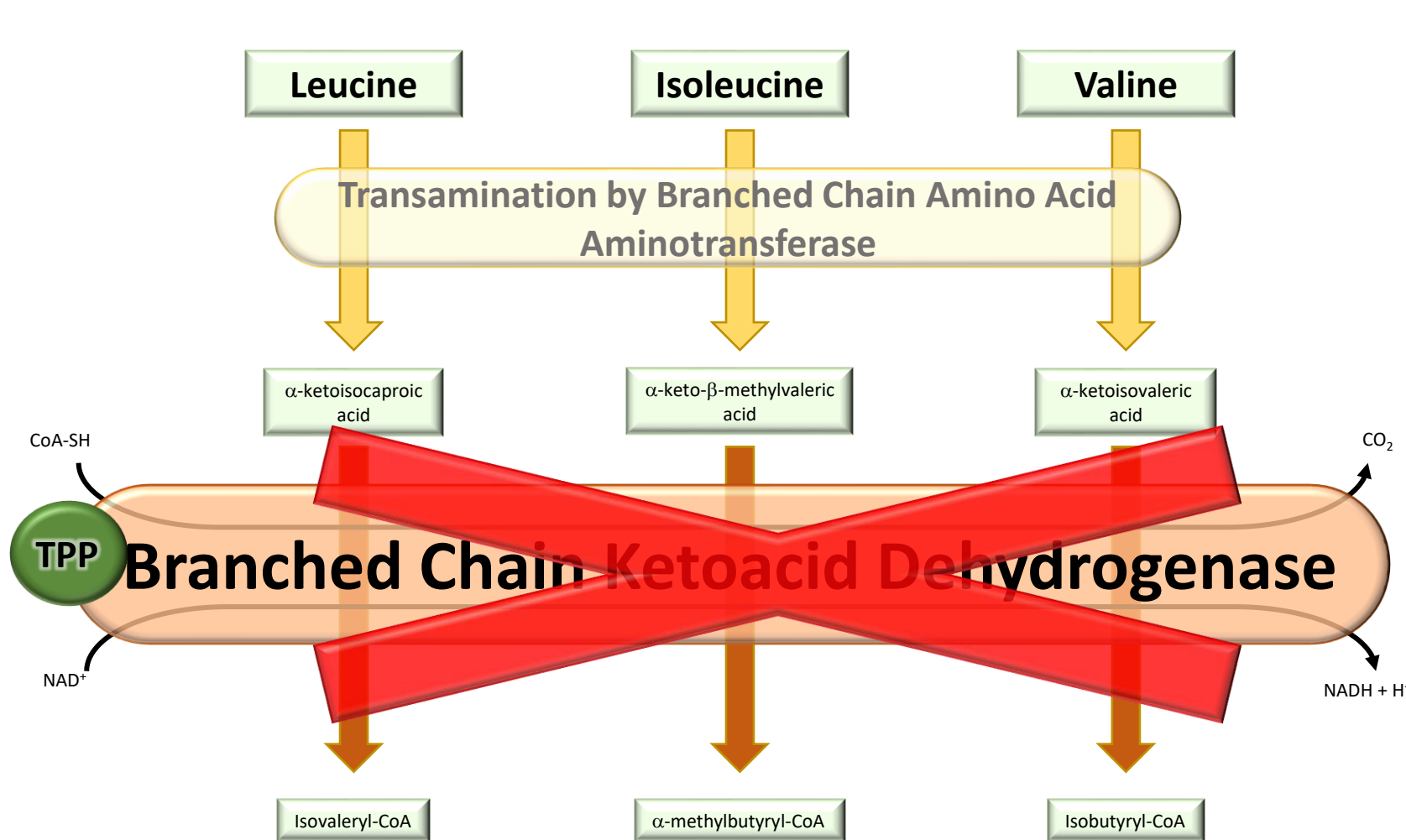


Teenager, ataxic, severe mental retardation



Frozen urine with concentrated oil at top with odor of maple syrup

Disorders in Metabolism of Branched-Chain Amino Acids



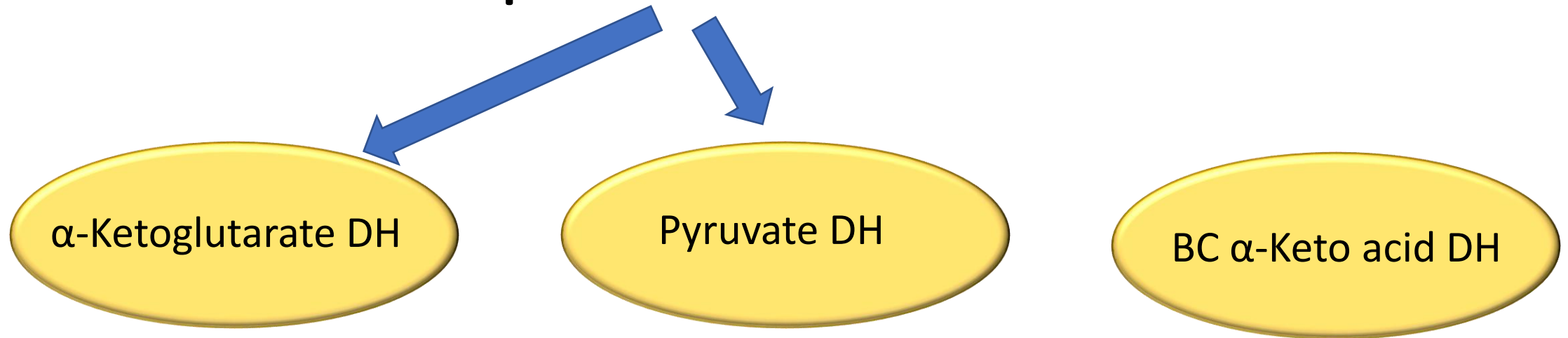
MSUD

- Defect in BCK Dehydrogenase Complex causes MSUD
- Increases in ketoacids are neurotoxic, particularly **α -ketoisocaproic acid**
- Ketoacid accumulation in urine responsible for maple syrup/burnt sugar odor
- What cofactors might improve residual enzyme activity?
 - **TPP**

Treatment of MSUD

- **Acute:** 20% glucose with insulin IV
- **Long-term:** Dietary restriction of branched chain amino acids (leucine, isoleucine, valine) improves the neurological manifestations.
 - Thiamine supplementation
 - Very difficult to treat; requires lifelong restriction
 - BCAA are very abundant in most protein sources
 - The 3 BCAA are all essential
 - Valine and isoleucine supplementation (recall α -ketoisocaproic acid is most toxic)
 - Avoidance of catabolic states: causes the mobilization of tissue protein which then releases amino acids
- With strict dietary adherence patients have lived to adulthood
<http://www.ncbi.nlm.nih.gov/pubmedhealth/PMH0001411>)

Link this concept: Recall



BC Keto acid Dehydrogenase and the other dehydrogenases have similar reactions, similar mechanisms, similar cofactors (all need Vit B1 (thiamine pyrophosphate):

Consider the importance of Vit B1 (thiamine pyrophosphate)

Methylmalonic Aciduria

Methylmalonic Aciduria Scenario

- Lara was born in a private hospital in Portugal
- The birth followed a normal pregnancy, but it became immediately apparent that Lara had **feeding problems, refusing to suck** and then **vomiting** after a bottle feed. However, this settled after a while and at four days we were allowed home.
- It was only after a few hours of being home that Lara developed **breathing problems** and we rushed her straight back to hospital where we were told that she was extremely **dehydrated**.
- Characteristic features were **metabolic acidosis, poor feeding & lethargy**.
- On admission, **methylmalonate levels** in circulation were **elevated**.

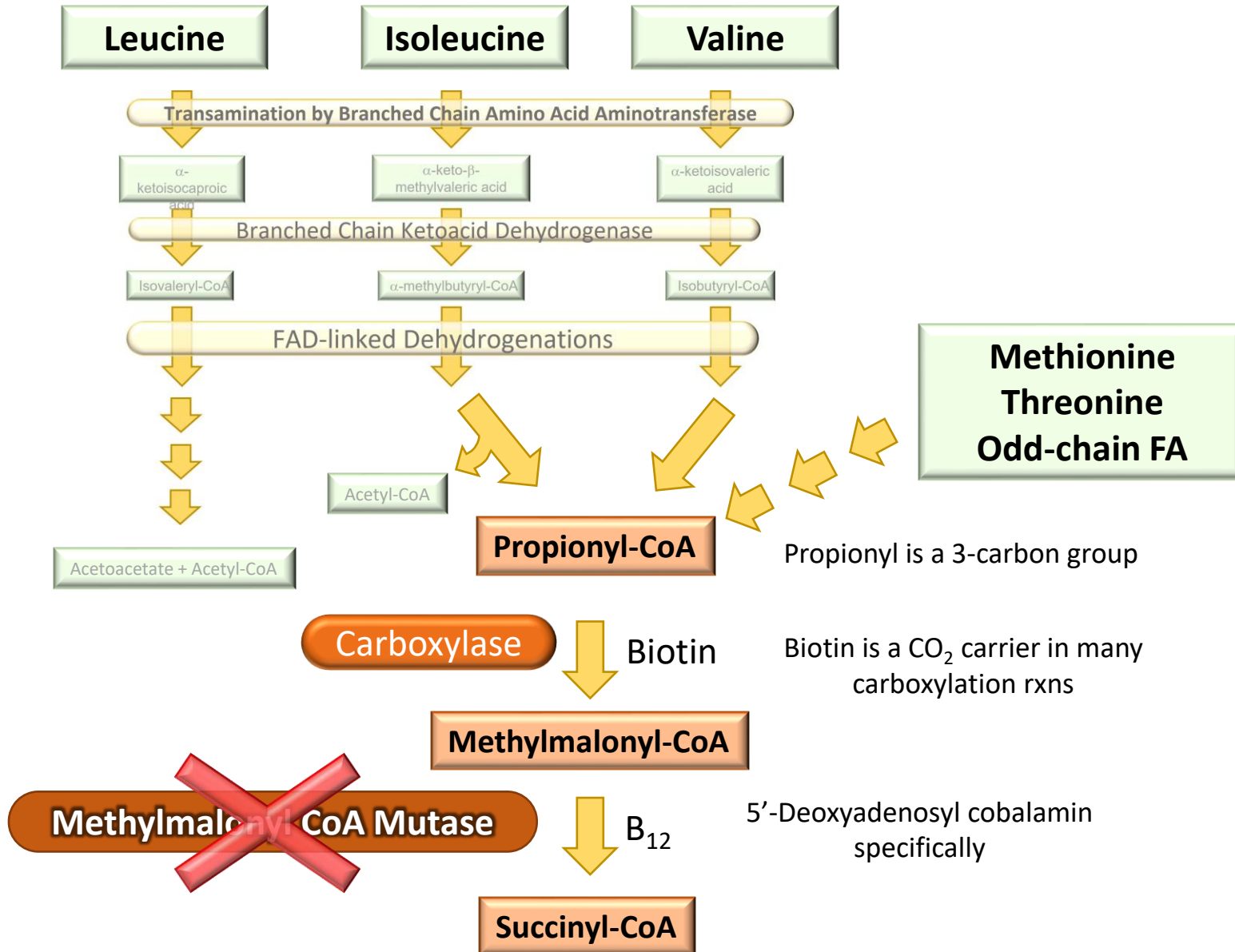
Methylmalonyl CoA Mutase Deficiency

- Results in elevated levels of methylmalonic acid in circulation, found by GC/MS analysis on urine
- Causes **metabolic acidosis** due to the accumulation of methylmalonic acid
 - Ketosis and ketoacidosis
- Associated with **neurological manifestations: seizures, encephalopathy**
- Severity of disease is typically related to the amount of enzymatic activity
 - Neonatal form
 - Chronic intermittent form
 - Chronic progressive form

} Details not required for this course



Methylmalonyl CoA mutase deficiency



MMA

Treatment:

- May show improvement with **vitamin B₁₂ (cobalamin)** supplementation if enzyme deficiency is due to a mutation in the B₁₂ binding site of methylmalonyl CoA mutase
- Consider diet with reduced Isoleucine, Valine, Methionine, Threonine

Homocysteine Metabolism

Homocystinuria

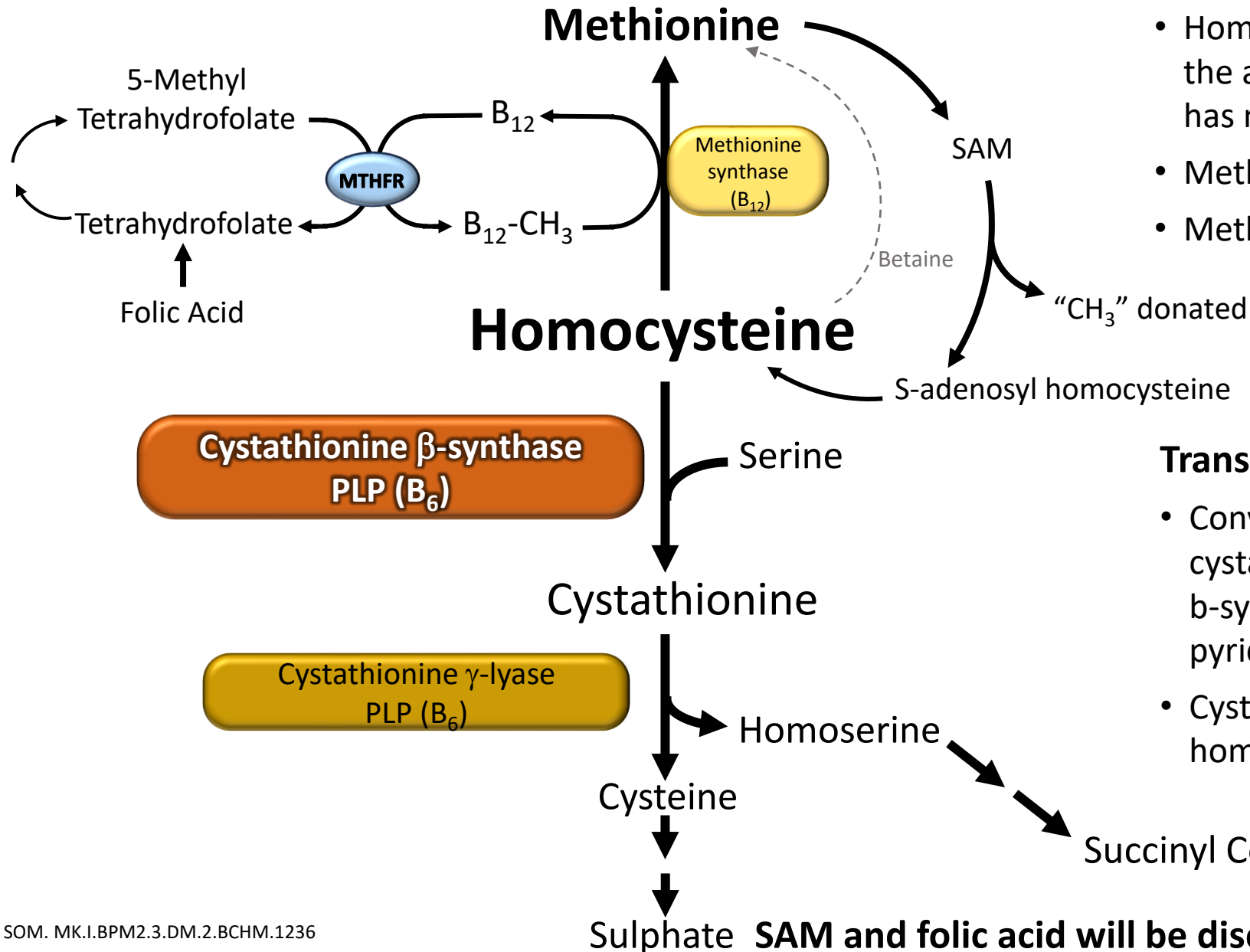
- A group of disorders in which there is a defect in homocysteine metabolism
- Characterized by high plasma & urinary levels of homocysteine
- Deficiency of *cystathionine β -synthase* (transulfuration pathway)
- Homocysteine binds to connective tissue and disrupts its structure
- Characterized by dislocation of lens (ectopia lentis), Skeletal abnormalities sometimes referred to as Marfanoid Habitus, Delay of mental development, premature arterial disease
- Some patients respond to oral vitamin B₆



Homocystinuria scenario

- As Kelly approached the ages of three and four, it became obvious that she was becoming more and more agitated and was losing fine motor skills. Just before starting kindergarten, she was diagnosed with **subluxed lenses in both eyes** and high myopia. After a thorough work up by a pediatric neurologist, Kelly was diagnosed with **Homocystinuria**.
- The treatment for the disorder seemed almost worse than the disease itself. Kelly was 6 years old and lived to eat. All her favorite foods were taken away and she was forced to take a nasty tasting **medical formula**.
- Last year, Kelly had a severe sinus infection. The MRI showed the affected sinus pushing on a cranial nerve, requiring immediate surgery.
- MRI also showed that her internal carotid **arteries were in the process of narrowing**.

Two Fates of Homocysteine



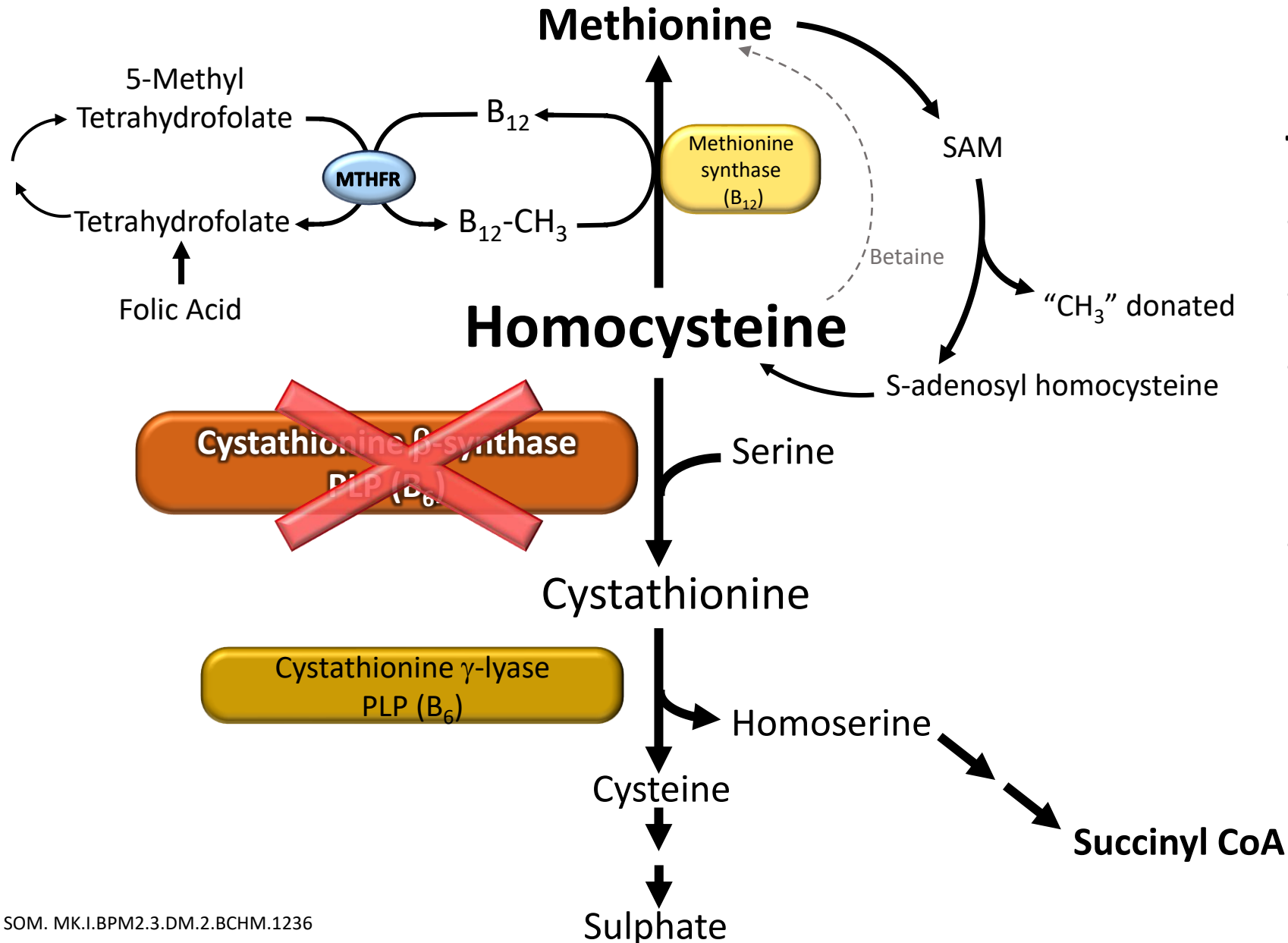
Methylation reactions

- Homocysteine is recycled to methionine by the activity of Methionine Synthase which has methylcobalamin as cofactor
- Methylcobalamin restored by 5-methyl THF
- Methyl group is donated to reactions

Transulfuration reactions

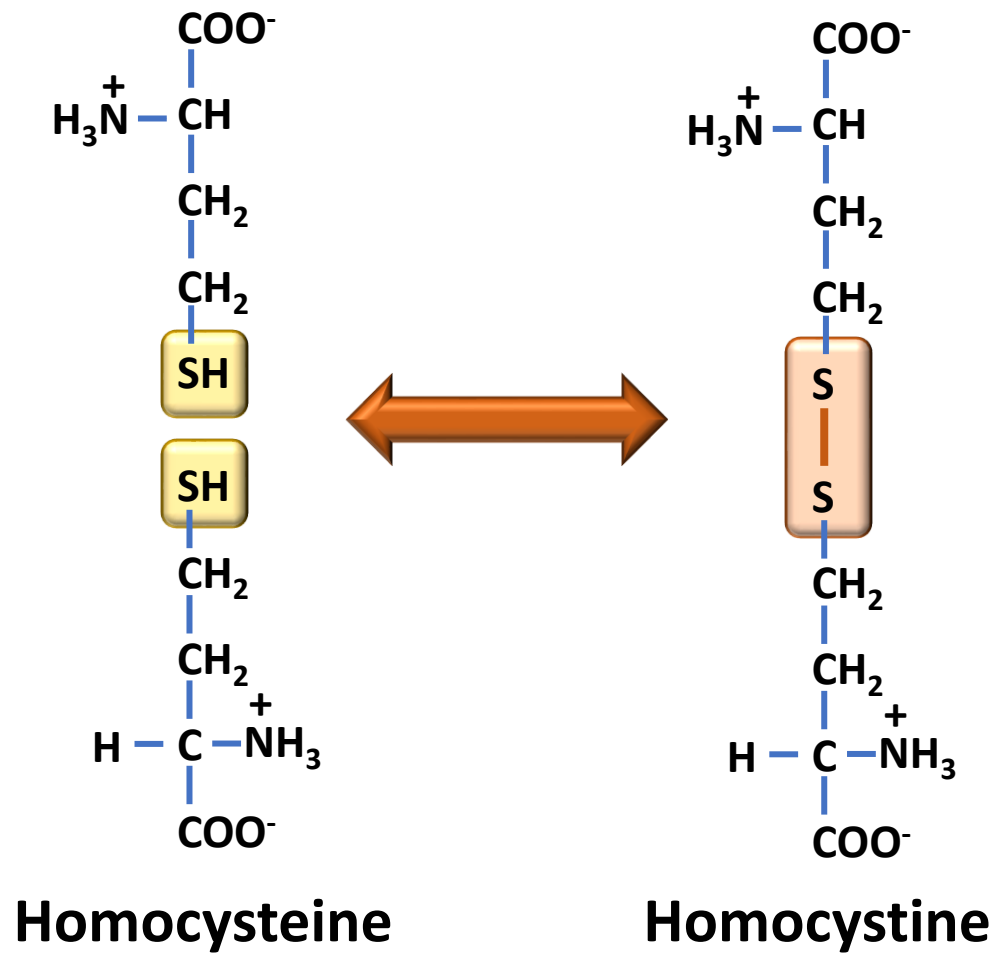
- Conversion of homocysteine to cystathionine by the activity cystathionine b-synthase (CBS) Synthase which has pyridoxal phosphate (PLP) as cofactor
- Cystathionine is converted to cysteine and homoserine, each is further metabolized

Homocystinuria Metabolic Defect



Therapy:

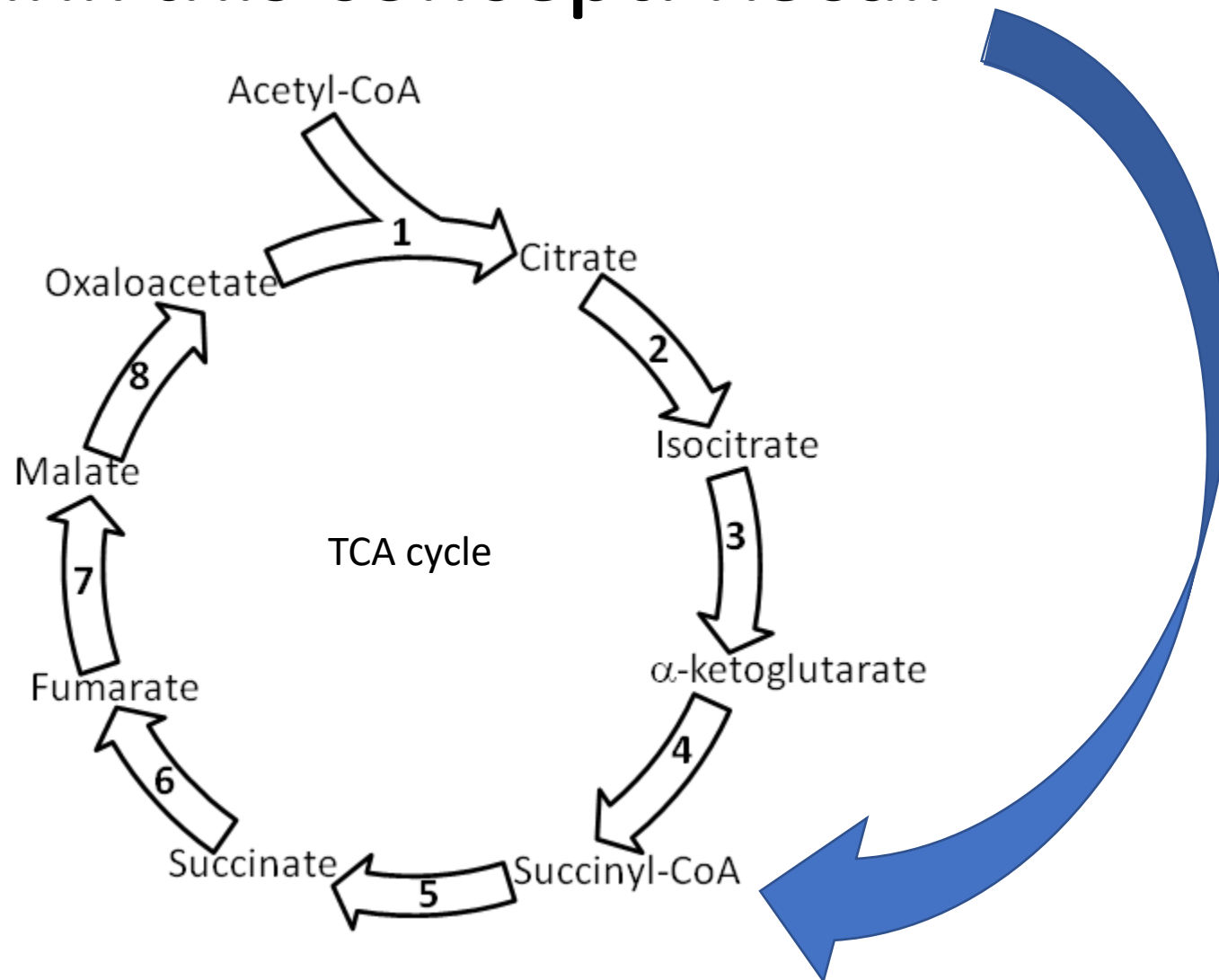
- Pyridoxine (B6)
- Restrict dietary protein
- Supplement with methionine free medical food
- Oral Betaine-HCl



Homocysteine Interferes with Collagen

- Leads to lens dislocation after age 3
 - other ocular abnormalities as well
- Osteoporosis develops during childhood, mental retardation
- Lipid deposits form atheromas
- Homocysteine can have other effects:
 - Lipid oxidation and platelet aggregation
 - leads to fibrosis
 - calcification of atherosclerotic plaques

Link this concept: Recall



Succinyl-CoA:

- Is an intermediate of TCA cycle
- Therefore **methionine**, **serine** and **cysteine** are glucogenic amino acids

Link this Concept:

Compare Homocystinuria to Marfan Syndrome

Homocystinuria	Marfan Syndrome
Tall stature with arachnodactyly	Tall stature with arachnodactyly
Increased arm to height ratio	Increased arm to height ratio
Scoliosis and/or pectus deformity	Scoliosis and/or pectus deformity
Autosomal Recessive	Autosomal Dominant
Limited joint mobility	Lax joint - hyperflexibility
Lens dislocation downward & in (toward nose) 	Lens dislocation upward (Marfan, ceiling fan is up) and out (toward temple) 
Low IQ	Normal IQ
Normal aorta	Aortic dilation

Disorders of Amino Acid Transport

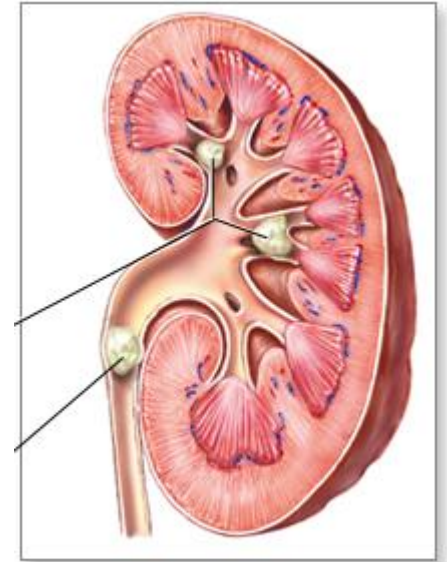
Cystinuria

Cystinuria

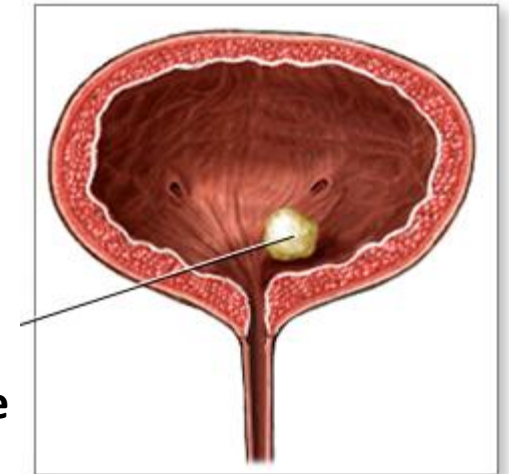
- Inherited deficiency of the Cystine transporter
- Defective Tubular reabsorption of dibasic amino acids:
Cystine, ornithine, arginine, lysine

COLA or COAL is the acronym

- Cystine is excreted in urine
- Cystine tends to precipitate in the renal tubules (cystine stones) in the renal tract
- One of the causes of renal stones in children
 - Cystine crystallizes at high concentrations
- Therapy: High fluid intake



ADAM.



ADAM.
57

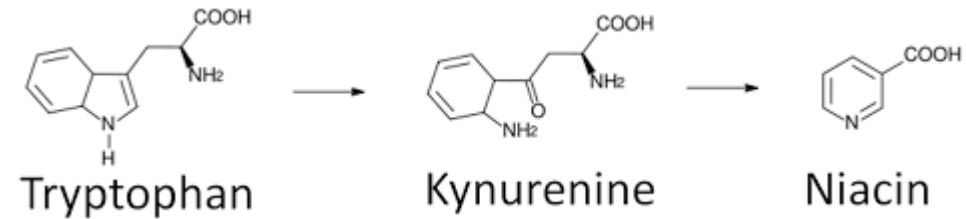


Hexagonal
crystals in urine

Disorders of Amino Acid Transport

Hartnup Disease

Hartnup Disease



- Reabsorption defect in neutral amino acids
 - Ala, val, ser, thr, leu, ile, phe, tyr, **trp**, his, gln, asn
 - High levels in urine
- Low Tryptophan concentrations are pathogenically relevant due to **decreased dietary absorption** and **increased excretion**
 - Most patients show no clinical features
- **If patient presents with clinical features;**
 - Photodermatitis, cerebellar ataxia
- Low tryptophan = low B3 (niacin) = low NAD⁺ = Pellagra
- 4 signs of Pellagra
 - Diarrhea, dermatitis, dementia, death
- Treat with high nicotinamide, protect from sun

Amino acid disorders

Enzyme

Co-enzyme

Clinical Features

Classic PKU			
Phenylketonuria (BH4 deficiency)			
Alkaptonuria			
MSUD			
Homocystinuria			
Cystinuria			
Hartnup's disease			

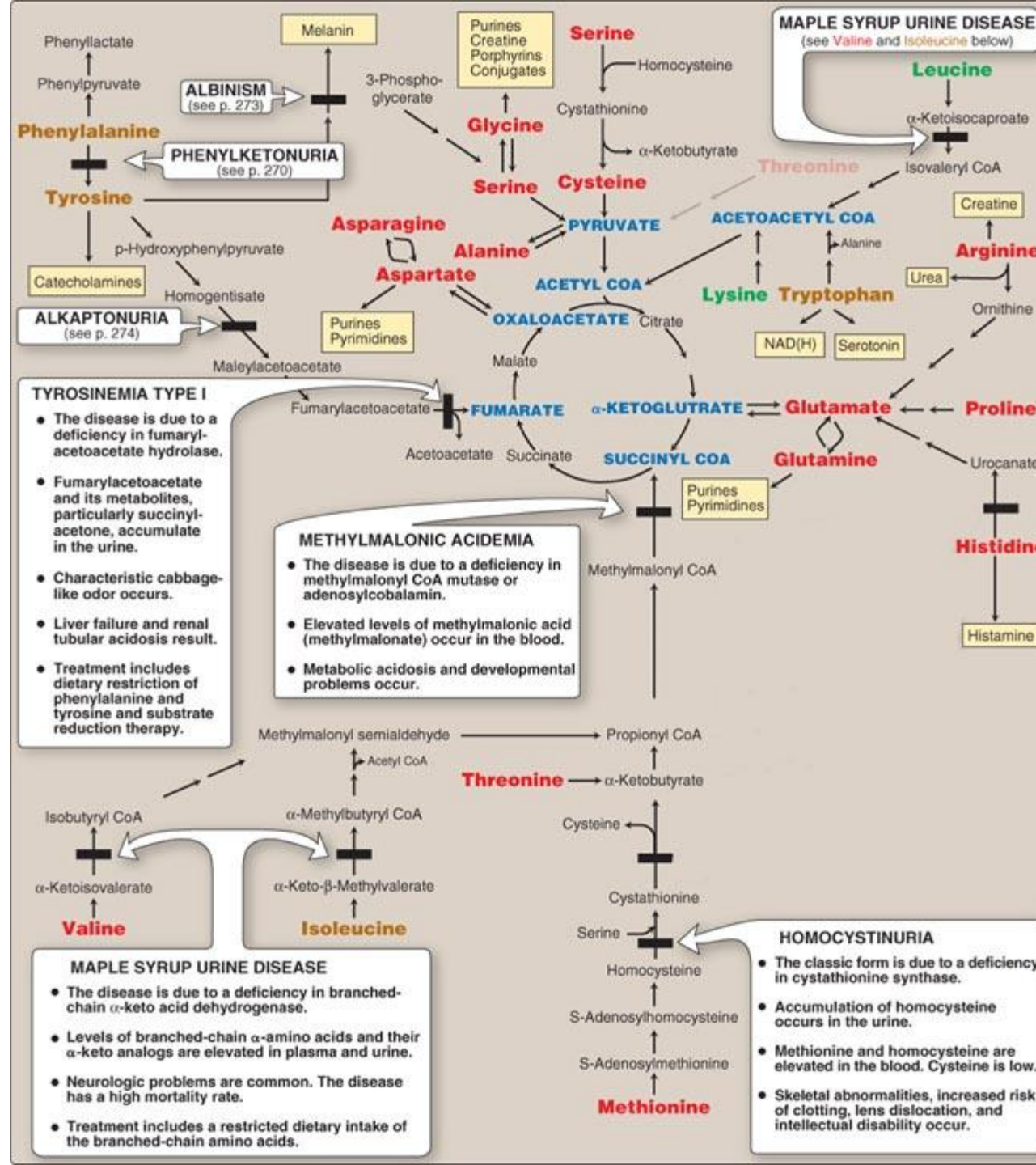
Amino acid disorders	Enzyme	Co-enzyme	Clinical Features
Classic PKU	Phenylalanine hydroxylase	Biopterin (BH₄)	Developmental delay, ↑Phe, ↓IQ, ↓pigmentation hair & skin, mousey urine smell
Phenylketonuria (Saproterin (BH ₄) deficiency)	Mainly Dihydropteridine reductase to restore BH ₂ to BH ₄	NADH	Developmental delay, ↑Phe, ↓IQ, ↓pigmentation hair & skin, mousey urine smell
Alkaptonuria	Homogentisic Acid Oxidase	No	↑Homogentisic acid, urine turns dark upon standing, accumulates in cartilage & connective tissue (collagenous), degenerative arthritis
MSUD	Branched Chain Ketoacid Dehydrogenase	TPP , Lipoic acid, CoA, FAD, NAD ⁺	↑valine, leucine & isoleucine, poor feeding & weight gain with vomiting, neurological deficits, ketoacidosis, maple or burnt sugar urine smell
Homocystinuria	Cystathionine β-Synthase	PLP	↑homocysteine, Disrupts structure of connective tissue, lens dislocation, skeletal abnormalities (Marfanoid Habitus), delay of mental development, premature arterial disease
Cystinuria	Dibasic amino acids transporter	no	Accumulation of cystine in urine, lysine, arginine and ornithine causing renal stones (COAL)
Hartnup's disease	Neutral amino acid transporter	no	↑ Neutral amino acids in urine, Photosensitive rashes, neurological abnormalities including ataxia and increased muscle tone (↑ala, ser, thr, val, leu, ild, phe, tyr, trp, his, gln, asn)

List some coenzymes important in amino acid metabolism and list at least one reaction requiring that coenzyme

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Methionine synthase	B 12; folic acid
Phenylalanine hydroxylase	BH4
Dopa decarboxylase	B6
Cystathione b-synthase	B6
Pyruvate dehydrogenase	B1;
Pyruvate carboxylase; Propionyl CoA carboxylase; Acetyl CoA carboxylase	Biotin
Methylmalonyl CoA mutase	B12

The End



Summary of the metabolism of amino acids in humans.

Genetically determined enzyme deficiencies are summarized in white boxes.

Nitrogen-containing compounds derived from amino acids are shown in small, yellow boxes.

Classification of amino acids is color coded:
Red = glucogenic;
brown = glucogenic and ketogenic;
green = ketogenic.

Compounds in BLUE ALL CAPS are the seven metabolites to which all amino acid metabolism converges.

CoA = coenzyme A;
 NAD(H)=nicotinamide adenine dinucleotide

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